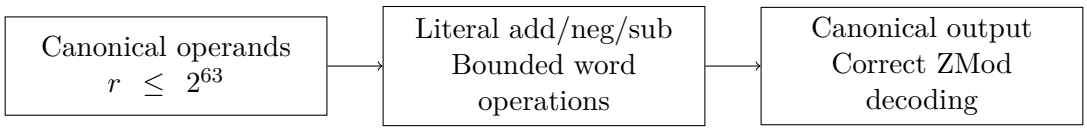
Independent graph Ring word review

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Seven main declarations, ten producer controls and nine reviewer controls passed fresh isolated Lean replay. Separate axiom readback covers all26 declarations with only standard foundations; two declarations are axiom-free. No discrepancy was found in the stated conditional value semantics.



Source and additive boundary. Native Ring uses $(a + b) \bmod r$, $(r - a) \bmod r$, and subtraction through negation then addition. Canonical $a, b < r \leq 2^{63}$ imply $a + b \leq 2^{64} - 2$. The endpoint is valid for the theorem. Ring::new checks primality and $r > 2$ but does not enforce this upper bound; constructor success alone cannot establish additive headroom.

Literal negation. The bridge to reduced-input negation is proved using $a \bmod r = a$ under canonicality. It does not silently replace the native expression outside that domain. Negation needs no small-modulus bound beyond positive UInt64 modulus and canonical operand.

Wide multiplication. Both operands widen to128bits before multiplying. The inspected Wide dependency proves the full product fits and remainder lies below $r < 2^{64}$, so narrowing is exact. Arbitrary UInt64 operands and any positive modulus are supported by this value model. Retained imported objects were hash-bound, not all rebuilt.

Signed branch. Native i8 sign==1 passes the operand through; every other sign negates it. The theorem uses Int sign restricted to $+1$ or -1. The i8 decoding bridge and graph-state sign preservation remain separate. Sign2 is not multiplication by2; sign1 does not canonicalize its input.

Independent controls. Nine kernel controls cover maximum additive headroom, endpoint subtraction, maximal-modulus negation, maximal operand products wide versus narrow, sign2, noncanonical sign1, literal negation underflow and modulus1. Out-of-bound UInt64 examples describe wrapping; Rust overflow-checking builds can panic instead.

Remaining scope. Inverse, prime-constructor correctness, counters/overflow, compiler equivalence and graph potential invariants are not established. Fresh Lean4.34.1 target builds reuse broader Mathlib caches. Private snapshots are hash-bound but not published commits, so full formal-review workflow completion remains false. Full1260-target scope and null cost ratio remain; no scientific execution or measurements occurred.

Source anchor: three_leads.rs.txt:1492–1534. See REVIEW.md, status.json, axiom-audit.json and exact source/dependency bindings. The first audit parser omitted two axiom-free lines; expected-count validation detected it, and the corrected parser verifies all26.